

Normal Distribution

Hisham A. Nematalla, PhD, PharmD.

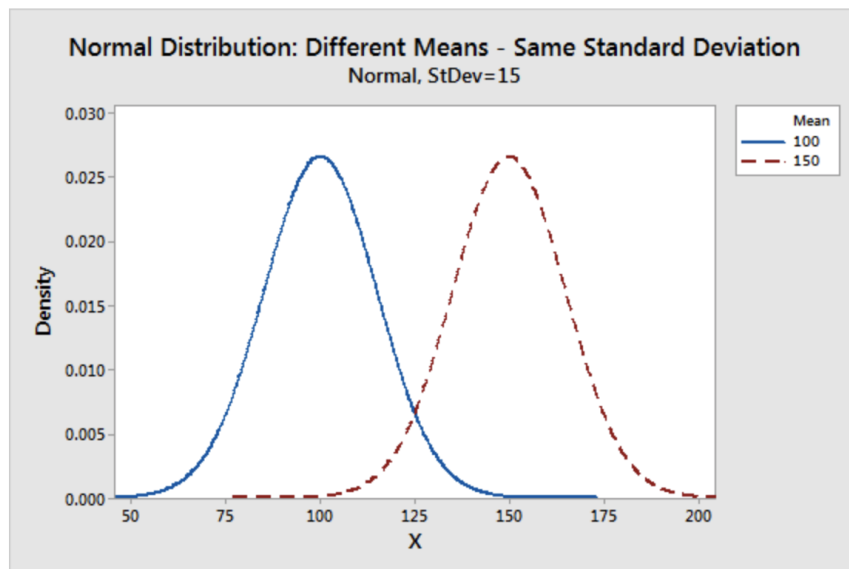
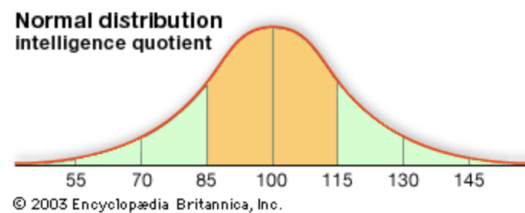
NORMAL DISTRIBUTION

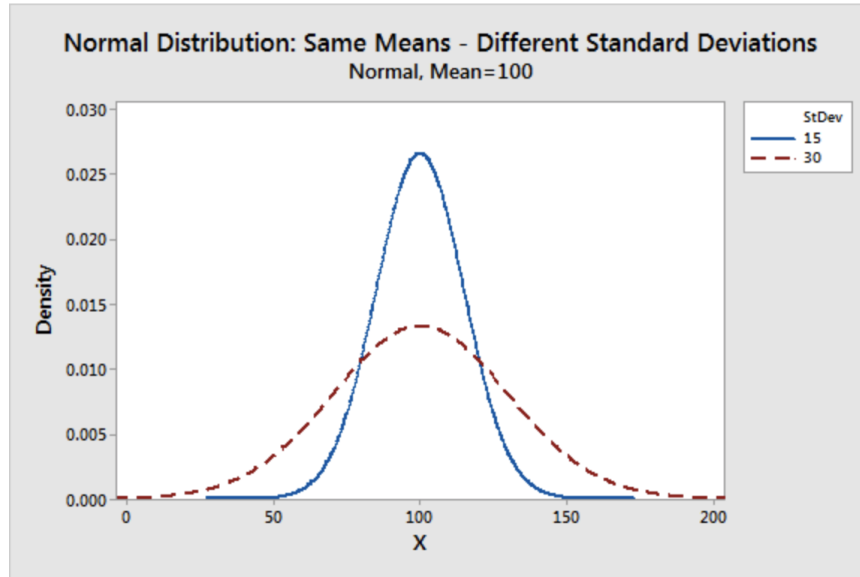
- The normal distribution is also called the **Gaussian distribution**.
- Named in honor of its discoverer Carl Friedrich Gauss, a very famous German mathematician.
- Normal distribution, there's an **infinite number of normal distributions**.
- They are defined by two quantities.
 - **Mean**
 - **Standard deviation**.



The graph of the normal distribution is characterized by two parameters:

- **Mean**, or average, which is the maximum of the graph and about which the graph is always symmetric;
- **Standard deviation**, which determines the amount of dispersion away from the mean.
- A **small standard deviation** (compared with the mean) produces a **steep graph**, whereas a **large standard deviation** (again compared with the mean) produces a **flat graph**.

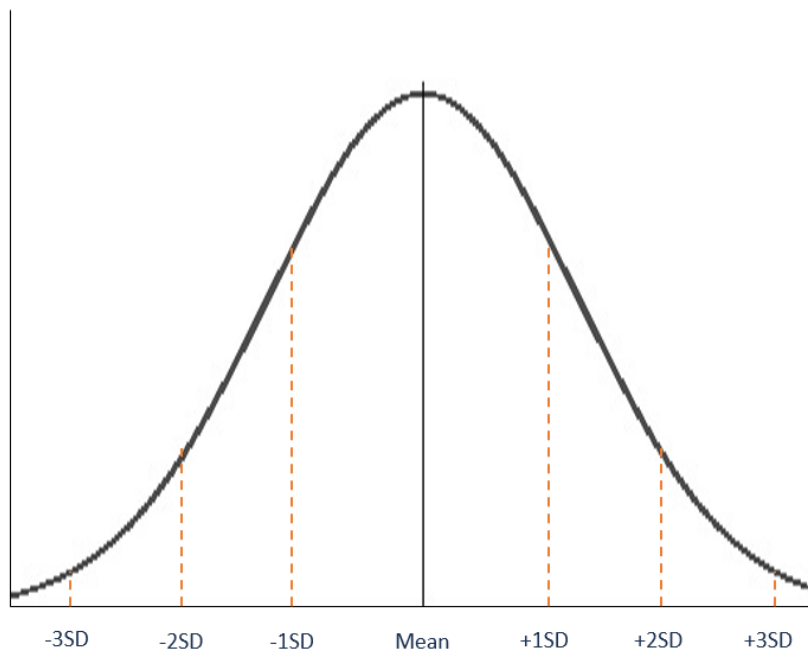
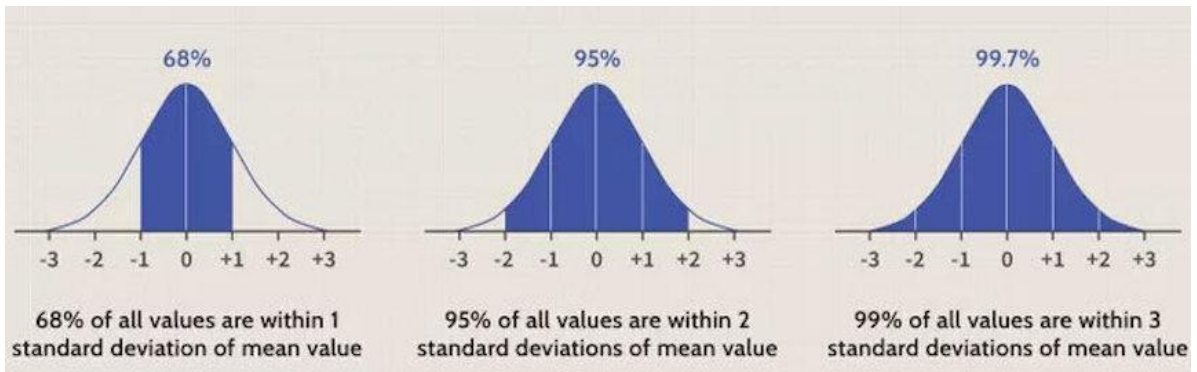




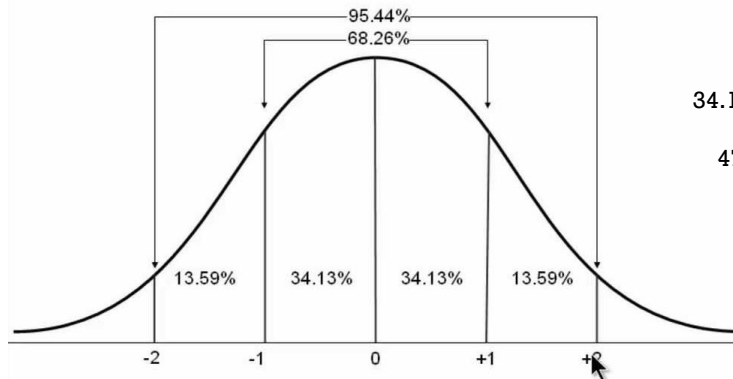
Despite the different shapes, all forms of the normal distribution have the following characteristic properties:

- They're all **symmetric bell curves**. The Gaussian distribution cannot model **skewed distributions**.
- The **mean, median, and mode are all equal**.
- Half of the population is **less than the mean and half is greater** than the mean.
- The **Empirical Rule** allows you to determine the **proportion of values that fall within certain distances from the mean**.





Normal Distribution



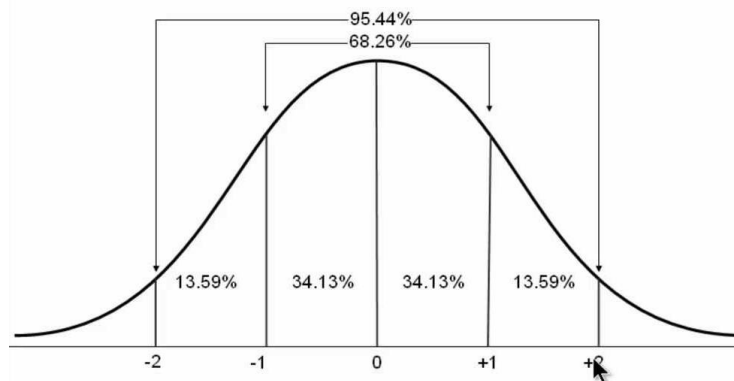
$$34.13 + 13.59 = 47.72 \%$$

$$47.72 \times 2 = 95.44 \%$$

- The probability that any randomly selected value from this distribution falls within **plus or minus one standard deviation of the mean is 68% (34.13 + 34.13 = 68.26 %)**.
- So if I pull a random observation from this distribution, there's a **68% chance that it's within plus or minus one standard deviations of the mean.**



Normal Distribution



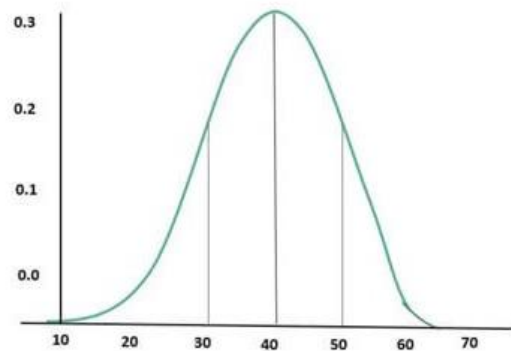
95% of the observations fall within two standard deviations of the mean (actually 1.96).



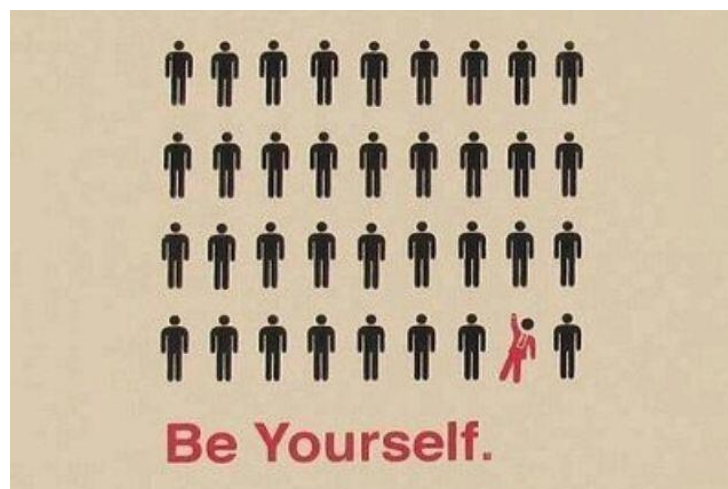
Why look at distribution?

- Patients often want to know whether their results on a lab test are average; if the results are not average, patients want to know **how far above or below average their results are**.
- Comparing **a score to a mean** is part of measuring relative location, typically defined as a score's position on the number line in relation to the mean.

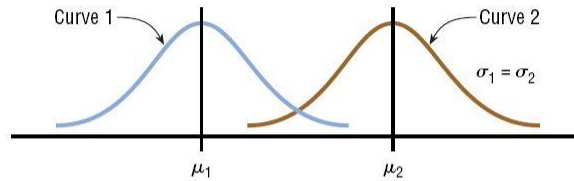
Normal Distribution



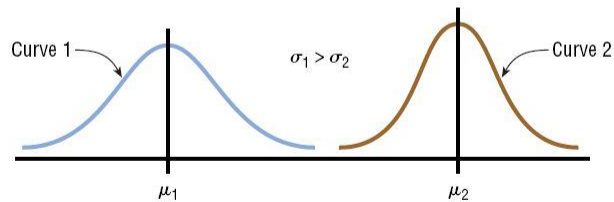
Where do I stand compared to the population (sample)?



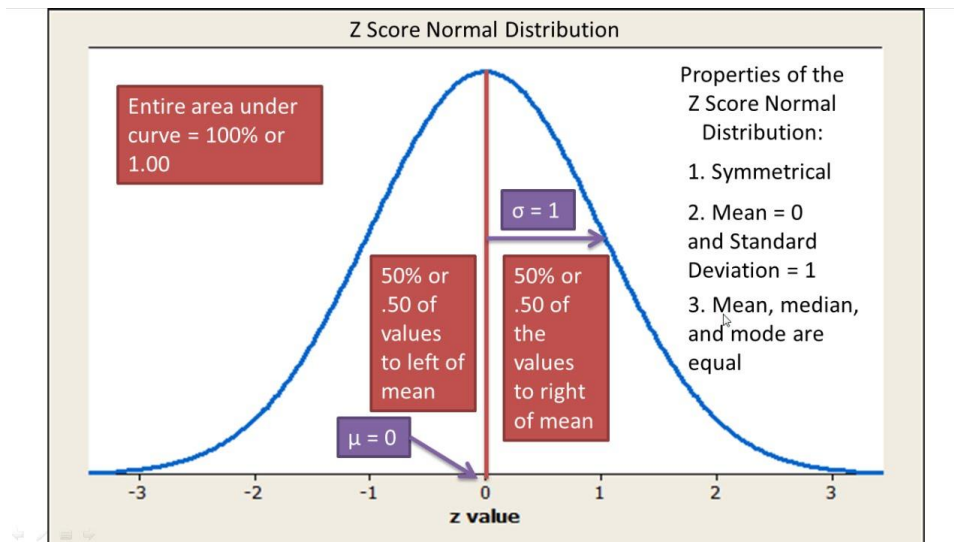
NORMAL DISTRIBUTIONS



(b) Different means but same standard deviations



(c) Different means and different standard deviations



- A **z score** is a mathematical calculation that measures the location of a **raw score relative to the mean**, with **distance measured in units of standard deviations**.
- The **z score** tells **whether the score is above or below the mean and how many standard deviations fit in the gap between the score and the mean**.
- Here is a **verbal definition of a z score**:
(Something minus its mean) divided by its standard deviation

$$Z = \frac{\text{raw score} - \text{mean}}{\text{standard deviation}}$$

Where

X is the raw score

μ is the mean

σ is the standard deviation

$$Z = \frac{X - \mu}{\sigma}$$

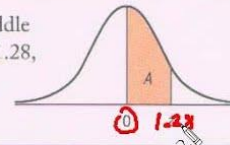


- A **z score** is also called a **standard score** because we are computing **the relative position from the mean in units of standard deviations**.
- Using a standard score **frees us from having to know the numeric value of a typical score on a possibly unfamiliar measure**.
- The **z comes** from the word **standardize**. When we standardize the data, we are obtaining scores that are in units of standard deviations.



Normal Curve (z) Table

The normal curve table gives only the percentage of data starting from the middle ($z = 0$), out to whatever z score you look up. For instance, if you look up $z = 1.28$, you get .3997. This means .3997 or 39.97% of the data in the normal curve is found between $z = 0$ and $z = 1.28$.

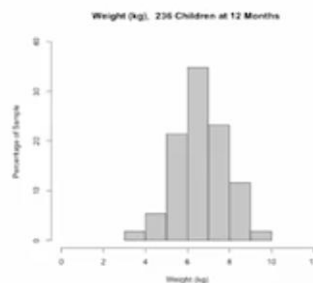


Normal										
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706

Example 1

- A basic histogram of the weight for 236 Nepali children at one year old

$$\begin{aligned}\bar{x} &= 7.1 \text{ kg} \\ s &= 1.2 \text{ kg} \\ \hat{m} &= 7.0 \text{ kg}\end{aligned}$$



- Suppose a mother brings her child to a pediatrician for the 12-month checkup, and wants to evaluate where the child's weight is relative to the population of 12-month-olds in Nepal
- Her child is 5 kg
- How does this child compare in weight to the weight of all 12-month-olds in Nepal?

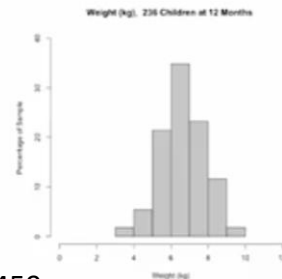
Example 1

- If we translate this measurement of 5 kg to units of standard deviation, we can find out where this child's weight compares to the mean of all such children

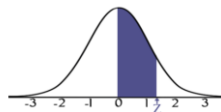
$$\text{Take } \frac{\text{individual observation} - \bar{x}}{s} = \frac{5 - 7.1}{1.2} = \frac{-2.1 \text{ kg}}{1.2 \text{ kg/s}} \approx -1.75 \text{ standard deviations}$$

- The original question ("How does my child's weight compare to the other children of the same age?") can be asked as, "What percentage of observations in a normal curve are more than 1.75 SDs below its mean?"

$$\begin{aligned}\bar{x} &= 7.1 \text{ kg} \\ s &= 1.2 \text{ kg} \\ \hat{m} &= 7.0 \text{ kg}\end{aligned}$$



Z score = -1.75 has a probability of 0.459.



STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and z standard deviations above the mean. For example, for z = 1.25 the area under the curve between the mean (0) and z is 0.3944.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
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0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
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1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

So 0.459 or **45.9%** of the data are between 0 and -1.75 on the standard normal distribution curve.

Example 2

In a class of students, the mean of grades was 80 and standard deviation was 3.

1. Calculate the z score?
2. What is the probability that a student gets 83 or less?
3. What is the probability that a student gets exactly 83?

$$\mu = 80$$

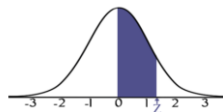
$$\sigma = 3$$

$$P(x \leq 83) = ?$$

$$Z = \frac{\text{raw score} - \text{mean}}{\text{standard deviation}}$$

$$Z = \frac{X - \mu}{\sigma}$$

$$Z \text{ score} = \frac{83 - 80}{3} = 1$$

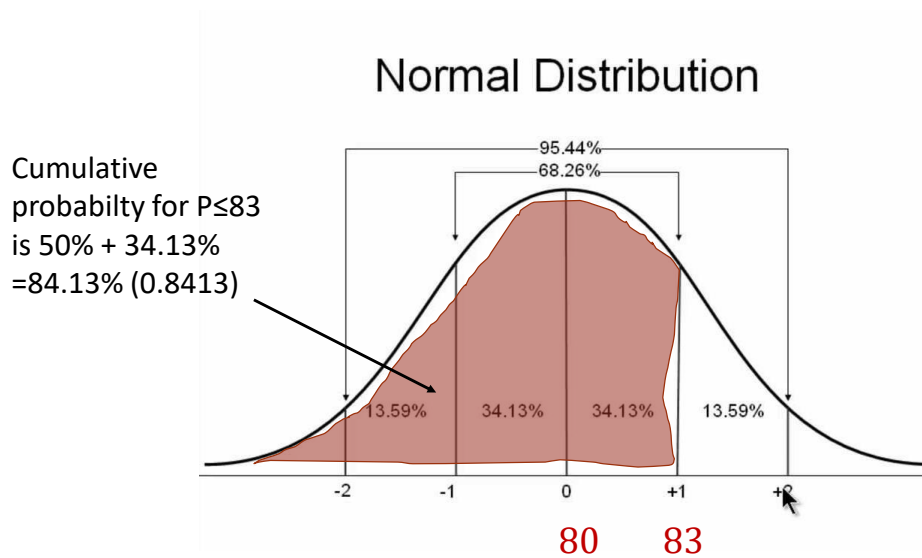


STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and z standard deviations above the mean. For example, for $z = 1.25$ the area under the curve between the mean (0) and z is 0.3944.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

So 0.3413 or **34.13%** of the data are between 0 and 1 on the standard normal distribution curve.

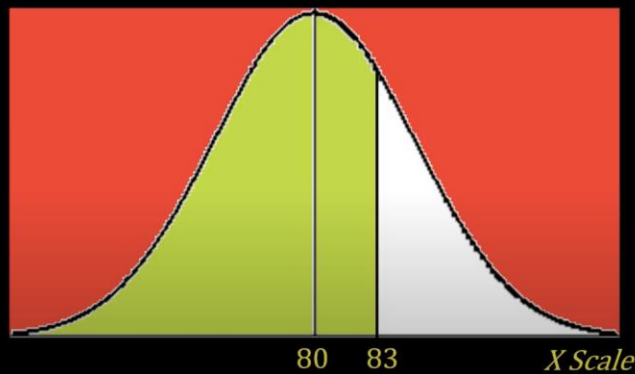


Example: Distribution of Grades

$$\mu = 80$$

$$P(X \leq 83)?$$

$$\sigma = 3$$

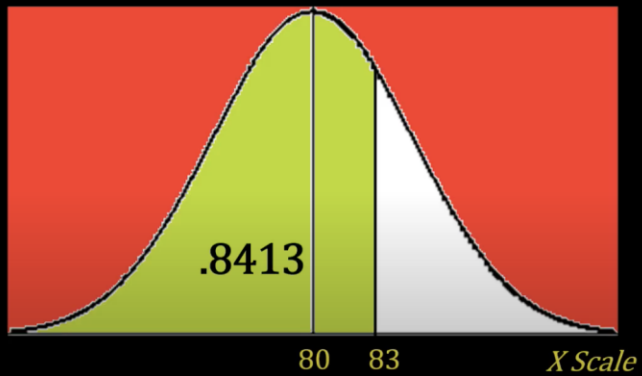


Example: Distribution of Grades

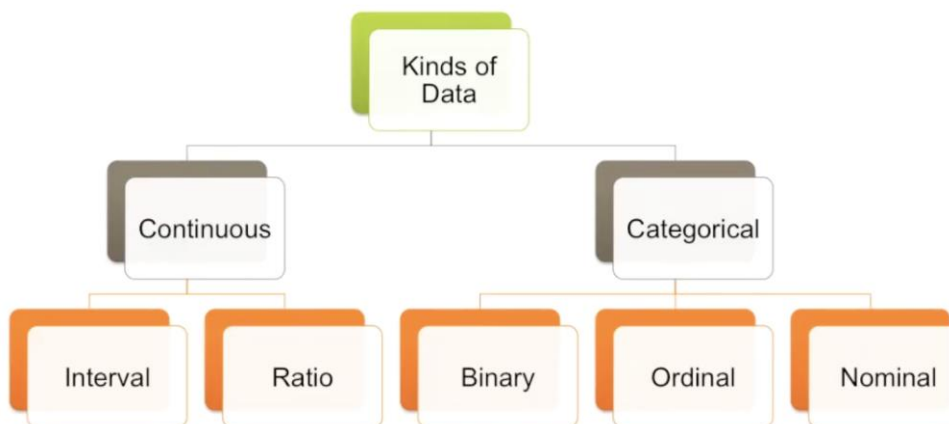
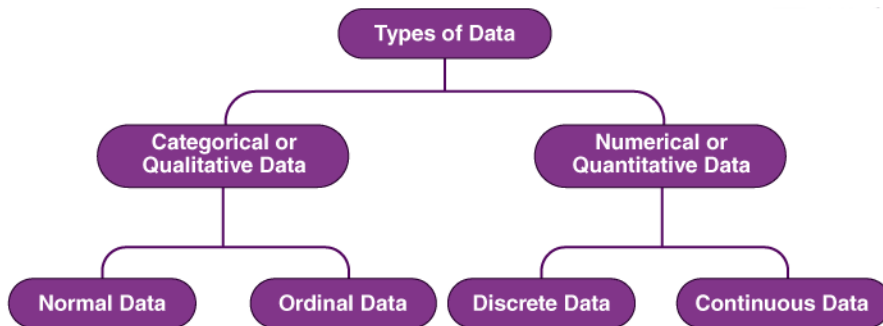
$$\mu = 80$$

$$\sigma = 3$$

$$P(X \leq 83) = .8413$$



BINARY DATA



Binary data occur when the **variable** of interest can only take two **values**. These two values are generally represented by 0 and 1, even if the variable is not quantitative.

Gender, the presence or absence of a characteristic and the success or failure of an experiment are just a few examples of **variables** that result in binary data. These variables are called **dichotomous variables**.

If there is an outcome it is **1**.
No outcome is **0**.



Encyclopedia of Statistics



Example

A meteorologist wants to know how reliable his forecasts are. To do this, he studies a **random variable** representing the prediction. This **variable** can only take two **values**:

$$X = \begin{cases} 0 & \text{if the prediction was incorrect} \\ 1 & \text{if the prediction was correct} \end{cases}$$

The meteorologist makes predictions for a period of **50** consecutive days. The prediction is found to be correct **32 times**, and incorrect **18 times**.



Descriptive Statistics

The only meaningful statistics would be the **proportion** of YES and NO to total number of sample.

True proportion in population (p)

Sample proportion ($\hat{p}$) "p-hat"



$$P = \text{Number of outcome} / \text{Total number}$$



The standard deviation of binary data is given by the following equation.

$$s = \sqrt{n\hat{p}(1 - \hat{p})}$$

However, unlike continuous data, it does not infer information on the distribution of the data.

- $\hat{p}$ summarizes the **percentage (probability, risk)** of outcomes among a sample.
- It also gives information about the **variability of individual** sample observations and the sample percentiles.
- $\hat{p}$ is the **sample mean** of sample observations that take the value 1 for observations with the outcome and 0 for observations without the outcome.



Example: Measure HIV status (yes/no) of babies born to mothers with HIV.

Randomly assign to 2 groups:

- one group receives AZT drug (n = 180)
- one group receive placebo (n = 183)

AZT: $\hat{p} = 13 / 180 = 7\%$

Placebo: $\hat{p} = 40 / 183 = 22\%$

	AZT	Placebo
HIV+	13	40
HIV-	167	143
Total	180	183

Connor EM, Sperling RS, Gelber R, Kiselev P, Scott G, O'Sullivan MJ, VanDyke R, Bey M, Shearer W, Jacobson RL, et al. Reduction of maternal-infant transmission of human immunodeficiency virus type 1 with zidovudine treatment. N Engl J Med. 1994 Nov 3;331(18):1173-80.



Summary Measures for Risk

Proportion of HIV+ in those having AZT treatment

$$p^{\wedge}_{AZT} = \frac{13}{180} = 0.07 \text{ or } 7\%$$

Proportion of HIV+ in those having placebo treatment

$$p^{\wedge}_{Placebo} = \frac{40}{183} = 0.22 \text{ or } 22\%$$

	AZT	Placebo	
HIV+	13	40	53
HIV-	167	143	310
Total	180	183	363

Connor EM, Sperling RS, Gelber R, Kiselev P, Scott G, O'Sullivan MJ, VanDyke R, Bey M, Shearer W, Jacobson RL, et al. Reduction of maternal-infant transmission of human immunodeficiency virus type 1 with zidovudine treatment. N Engl J Med. 1994 Nov 3;331(18):1173-80.



$$p^{\wedge}_{AZT} = \frac{13}{180} = 0.07 \text{ or } 7\% \quad \& \quad p^{\wedge}_{Placebo} = \frac{40}{183} = 0.22 \text{ or } 22\%$$

Risk difference (Attributable risk) = $p^{\wedge}_{AZT} - p^{\wedge}_{Placebo}$
 = $0.07 - 0.22 = -0.15$ (15% reduction)

Interpretation:

- There is **15% reduction in HIV+ transmission** to children born to mothers given AZT as compared to children born to mothers given placebo.
- There is **15% lower absolute risk of HIV+ transmission** to children born to mothers given AZT.

	AZT	Placebo	
HIV+	13	40	53
HIV-	167	143	310
Total	180	183	363

Connor EM, Sperling RS, Gelber R, Kiselev P, Scott G, O'Sullivan MJ, VanDyke R, Bey M, Shearer W, Jacobson RL, et al. Reduction of maternal-infant transmission of human immunodeficiency virus type 1 with zidovudine treatment. N Engl J Med. 1994 Nov 3;331(18):1173-80.



$$p^{\wedge}_{AZT} = \frac{13}{180} = 0.07 \text{ or } 7\% \quad \& \quad p^{\wedge}_{Placebo} = \frac{40}{183} = 0.22 \text{ or } 22\%$$

Risk ratio (relative risk) = $p^{\wedge}_{AZT} / p^{\wedge}_{Placebo}$
 = $0.07 / 0.22 = 0.32$ (32%)

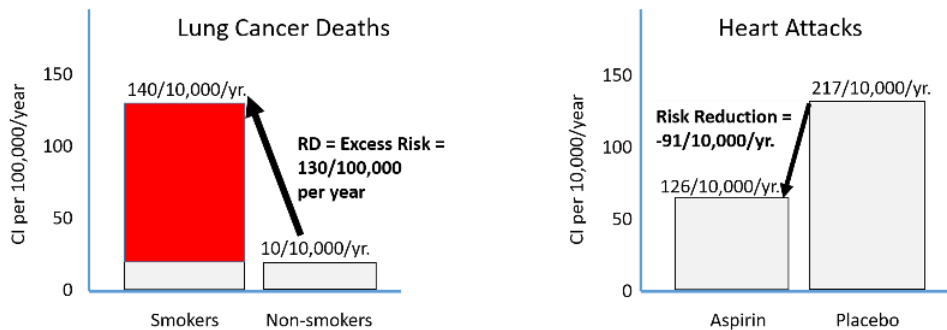
Interpretation:

- The risk that an **HIV+ mother who takes AZT during pregnancy transmits HIV to her child is 0.32 times her risk if she did not take AZT.**
- **The risk that an HIV+ mother transmits HIV to her child is 68% lower if she takes AZT during pregnancy.**

	AZT	Placebo	
HIV+	13	40	53
HIV-	167	143	310
Total	180	183	363

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Both risk difference and relative risk (risk ratio) give same information but differ in magnitude.



Exercise

Example: Hormone Replacement Therapy: Proportion of Women Developing CHD (Incidence), in a 2x2 Table

	HRT	PLACEBO	
CHD	163	122	285
No CHD	8,345	7,980	16,325
	8,508	8,102	16,610

$$\hat{p}_{HRT} = \frac{163}{8,508} \approx 0.019 \text{ (1.9\%)}, \quad \hat{p}_{PLACEBO} = \frac{122}{8,102} \approx 0.015 \text{ (1.5\%)}$$

$$\widehat{RD} = \hat{p}_{HRT} - \hat{p}_{PLACEBO} = 0.019 - 0.015 = 0.004 \text{ (0.4\%)}$$

$$\widehat{RR} = \frac{\hat{p}_{HRT}}{\hat{p}_{PLACEBO}} = \frac{0.019}{0.014} = 1.27$$



Summary

- ▶ Risk difference ($\hat{p}_1 - \hat{p}_2$) and relative risk ($\frac{\hat{p}_1}{\hat{p}_2}$) are two different estimates of the magnitude and direction of association for binary outcomes (between groups)
 - ▶ These two estimates are based on the exact same inputs and will always agree in terms of the direction of association, but not necessarily magnitude
- ▶ The *risk difference* helps to quantify the potential impact of a treatment or exposure for a group of individuals
- ▶ The *relative risk* helps quantify the potential impact of a treatment or exposure for an individual
- ▶ Neither estimate alone is sufficient to tell the whole “story”



For the given dataset, there are 3 summary measures for the outcome:

1. Risk difference (difference in proportions or attributable risk)

$$\hat{p}_{AZT} - \hat{p}_{Placebo} = 0.07 - 0.22 = -0.15 \text{ (15\% reduction)}$$

2. Relative risk (ratio of proportions or Risk ratio)

$$\frac{\hat{p}_{AZT}}{\hat{p}_{Placebo}} = \frac{0.07}{0.22} = 0.32$$

3. Odds ratio (relative odds)

	AZT	Placebo	
HIV+	13	40	53
HIV-	167	143	310
Total	180	183	363

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Odds ratio

What does Odds mean?

- The (estimated) odds of an event for any group is the (estimated) **probability of the event occurring divided by the (estimated) probability of it not occurring:**

$$\text{ODDS} = \frac{p^{\wedge}}{1-p^{\wedge}}$$

As $p^{\wedge}$ increases, ODDS increases.



AZT

Probability of the event occurring: $p^{\wedge}_{AZT} = 13/180 = 0.07$

Probability of it not occurring: $1-0.07$ OR $167/180 = 0.93$

$p^{\wedge}_{AZT} = 0.07$ so $\text{ODDS} = 0.07/(1-0.07) = 0.075$

Placebo

Probability of the event occurring: $p^{\wedge}_{Placebo} = 40/183 = 0.22$

Probability of it not occurring: $1-0.22$ OR $143/183 = 0.78$

$p^{\wedge}_{Placebo} = 0.22$ so $\text{ODDS} = 0.22/(1-0.22) = 0.28$

So

$$\text{Odds ratio} = \frac{\text{ODDS}_{AZT}}{\text{ODDS}_{Placebo}} = \frac{0.075}{0.28} = 0.26$$

This means that patients receiving AZT have odds **0.26 times** less of manifesting HIV+.

	AZT	Placebo	
HIV+	13	40	53
HIV-	167	143	310
Total	180	183	363



Odds Ratio

Odds ratio is used to find the probability of an outcome of an event when there are two possible outcomes and there is plausible causal effect.

For example, it can be used to measure the probability of a health repercussion based on whether or not the patient has been exposed to a certain substance. Or to find the probability of default when the customer income is high versus when it is low.

The function is defined by the following formula:

$$\text{Odds Ratio} = \frac{P_x / (1 - P_x)}{P_y / (1 - P_y)}$$



Relative risk vs Odds ratio

Both are relative measures of comparison, and neither imparts information about the absolute magnitude of the risk/odds in the groups being compared

Both measures use exact same information but can give numerically different results: both will always agree in terms of direction of association

If $\widehat{RR} > 1$, then $\widehat{OR} > 1$.

If $\widehat{RR} < 1$, then $\widehat{OR} < 1$.

If $\widehat{RR} = 1$, then $\widehat{OR} = 1$.

The smaller $\hat{p}_1$ and $\hat{p}_2$ are in value, the closer $\widehat{RR}$ and $\widehat{OR}$ will be in value



The odds ratio, $\widehat{OR}$, provides an alternative to the relative risk, $\widehat{RR}$, for quantifying the association between a binary outcome between groups

- The odds ratio is ratio of odds between two groups: odds is related to risk (probability, proportion)

The odds ratio and relative risk both estimate the association between a binary outcome between groups at the individual level

- These two measures will agree in terms of direction but not always magnitude
- The smaller the risk in the groups being compared, the more similar in value $\widehat{RR}$ and $\widehat{OR}$ are



Exercise

Example: Hormone Replacement Therapy: Proportion of Women Developing CHD (Incidence), in a 2x2 Table

	HRT	PLACEBO	
CHD	163	122	285
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$$\begin{aligned} \widehat{RD} &= \hat{p}_{HRT} - \hat{p}_{PLACEBO} = 0.019 - 0.015 = 0.004 \text{ (0.4\%)} \\ \widehat{RR} &= \frac{\hat{p}_{HRT}}{\hat{p}_{PLACEBO}} = \frac{0.019}{0.014} = 1.27 \end{aligned}$$



Calculating the Odds Ratio (OR)

	Disease (Case)	No Disease (Control)
Exposed	A	B
Unexposed	C	D

$$\text{OR} = \frac{\text{Odds that a case was exposed (A/C)}}{\text{Odds that a control was exposed (B/D)}} = \frac{AD}{BC}$$

	Relative risk		Odds ratio	
	Disease	No disease	Disease	No disease
Exposed	A	B	A'	B'
Unexposed	C	D	C'	D'
	$\frac{A/(A+B)}{C/(C+D)}$		$\frac{A'D'}{C'B'}$	

The estimation of relative risk provided by the odds ratio can be illustrated algebraically. In uncommon diseases $A' \ll B'$ and $C' \ll D'$. Thus $A'/(A' + B')$ is approximately A'/B' and $C'/(C' + D')$ is approximately C'/D' . Therefore, the risk in exposed versus unexposed is approximately $(A'/B')/(C'/D')$, or rearranged $(A'D')/(C'B')$

